

Dice Combinations

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to count the number of ways to construct sum n by throwing a dice one or more times. Each throw produces an outcome between 1 and 6.

For example, if $n = 3$, there are 4 ways:

- $1 + 1 + 1$
- $1 + 2$
- $2 + 1$
- 3

Input

The only input line has an integer n .

Output

Print the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^6$

Example

Input	Output
3	4